

Subject Code	AF5644
Subject Title	Advanced Topics in Digital Economics
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	None
Objectives	This subject acts as a crucial component, contributing significantly to the Programme Learning Outcomes. It supports Learning Outcome 2 by providing students with opportunities to learn and apply modern techniques and methods for analyzing issues related to the digital economy.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. comprehend the fundamental concepts in digital economics; b. master the basic digital economics theories; c. obtain economics/finance/accounting knowledge to understand and evaluate digital economic activities; d. become familiar with the latest applications of digital technologies in economics;
Subject Synopsis/ Indicative Syllabus	● Theory: The rise of digital econ; Historical perspective: Examples; driving factors. ● Applications: Two-sided platform; Big data and behavior-based pricing; Cryptocurrency; Digital payment and digital currency; Crowdfunding; Digital banking; Algorithmic trading and High Frequency Trading. ● Special Issues Technology and jobs; Surviving digital technology disruption; Relationship between technology and human; Firms in the digital era; Equity and fairness; Digital divide; Algorithmic aversion; Algorithm and discrimination; Algorithmic bias; Regulatory implications
Teaching/Learning Methodology	This subject will be delivered primarily through lectures/seminars for the discussion of economic concepts and their applications to the topics concerned. Active class participation is strongly encouraged. Other learning activities, such as presentation and quiz, will be flexibly incorporated in classes to help students achieving the learning outcomes.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d		
	1. Class participation	10%	✓	✓	✓	✓		
	2. Presentation and Written Report	40%	✓	✓	✓	✓		
	3. Final report	50%	✓	✓	✓	✓		
	Total	100 %						
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Class participation – Students have to participate actively in class and give their ideas to the in-class exercise. Presentation and Written Report– Students have to work on a project on group basis. They are required to submit a written report and make presentation on their findings. Final report – Each student need to apply the knowledge learned to design their own digital economic product or service, and submit an individual written report.								
Student Study Effort Expected	Class contact:							
	▪ Lectures / Seminars					39 Hrs.		
	Other student study effort:							
	▪ Self study and read course material					39 Hrs.		
	▪ Preparation for presentation					50 Hrs.		
	Total student study effort					128 Hrs.		

Reading List and References

- Avi Goldfarb and Catherine Tucker (2019). Digital Economics. *Journal of Economic Literature* 2019, 57(1), 3–43.
- Philippon, T. (2016). *The Fintech Opportunity* (No. w22476). National Bureau of Economic Research.
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- Foley, S., Karlsen, J. R., & Putniņš, T. J. (2019). Sex, drugs, and bitcoin: How much illegal activity is financed through cryptocurrencies?. *The Review of Financial Studies*, 32(5), 1798-1853.
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- Jack, W., & Suri, T. (2014). Risk sharing and transactions costs: Evidence from Kenya's mobile money revolution. *American Economic Review*, 104(1), 183-223.
- He, Q., & Li, X. (2021). The failure of Chinese peer-to-peer lending platforms: Finance and politics. *Journal of Corporate Finance*, 66, 101852.
- Choi, H. S., & Loh, R. K. (2023). Physical frictions and digital banking adoption. *Management Science*, Forthcoming.
- Kirilenko, A. A., & Lo, A. W. (2013). Moore's law versus murphy's law: Algorithmic trading and its discontents. *Journal of Economic Perspectives*, 27(2), 51-72.
- Eisenmann, T., Parker, G., and Van Alstyne, M. 2006, “Strategies for Two-Sided Markets”, Harvard Business Review.
- Rysman, M. (2009). The economics of two-sided markets. *Journal of Economic Perspectives*, 23(3), 125-143.
- Townley, C., Morrison, E., & Yeung, K. (2017). Big data and personalized price discrimination in EU competition law. *Yearbook of European Law*, 36, 683-748.
- Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6), 2188-2244.
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- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2018). Overcoming algorithm aversion: People will use imperfect algorithms if they can (even slightly) modify them. *Management Science*, 64(3), 1155-1170.
- Fuster, A., Goldsmith-Pinkham, P., Ramadorai, T., & Walther, A. (2022). Predictably unequal? The effects of machine learning on credit markets. *The Journal of Finance*, 77(1), 5-47.